

Name: _____

Roll Number: _____

Quiz-1

Max. Time: 12 mins

Max. Marks: 10

Note: Solve all parts. Limit your written responses to the provided space.

Q.1. [5] Choose by putting a check mark on the most appropriate option. Note: No cutting/overwriting allowed.

i. Number of nodes explored by DFID are always greater than that for DFS.

(A) True (B) False

ii. The path returned by A* algorithm on a graph is optimal only while using a:

(A) Admissible (B) Monotone (C) Both 'A' and 'B'

iii. If $g(n)$ varies from node to node in a graph of a problem then the A* algorithm:

(A) Is optimal (B) The same as Dijkstra's algorithm (C) Must use an admissible heuristic (D) All of the given choices

iv. _____ is the algorithm of choice while searching for a goal in uninformed way.

(A) DFS (B) BFS (C) DFID (D) A*

v. A _____ heuristic expands more states than a(n) _____ heuristic.

(A) Admissible, Informed (B) More Informed, less informed (C) Informed, Admissible (D) None of the given choices

Q.2. [2]

a) Define monotonicity of a heuristic function.

A heuristic function h is monotone if

1. For all states n_i and n_j , where n_j is a descendant of n_i ,

$$h(n_i) - h(n_j) \leq \text{cost}(n_i, n_j),$$

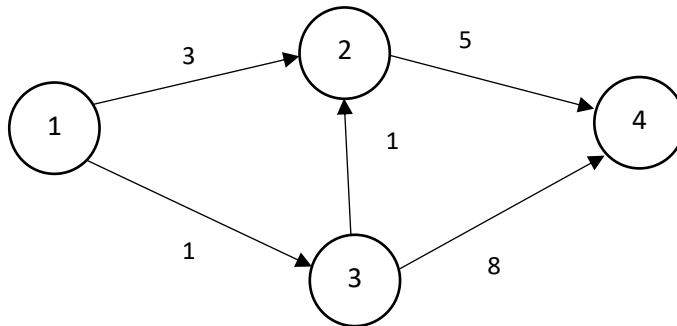
where $\text{cost}(n_i, n_j)$ is the actual cost (in number of moves) of going from state n_i to n_j .

2. The heuristic evaluation of the goal state is zero, or $h(\text{Goal}) = 0$.

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Q.3. [3] For the following directed graph, run the A* algorithm using the given heuristic function and state whether the heuristic is monotone or not. Show the contents of the OPEN and CLOSED lists using the hyphenated notation, e.g., for state 1, use 1-5, where ‘1’ is the state and ‘5’ is the heuristic estimate. Note that state ‘4’ is the goal state.



State	Heuristic Value
1	5
2	1
3	6
4	0

Solution: The given heuristic is not monotone because for states ‘1’ and ‘2’, the following inequality doesn’t hold.

$$h(1) - h(2) = 5 - 1 = 4 \not\leq cost(1,2) = 3$$

The result of running A* algorithm is shown in the table below.

<u>OPEN</u>	<u>CLOSED</u>
[1-5]	[]
[2-4,3-7]	[1-5]
[3-7,4-8]	[1-5,2-4]
[2-3,4-8]	[1-5,3-7]
[4-8]	[1-5,3-7,2-3]
Goal Found!	

Solution path: 1->3->2->4.